

Ex.No:1a	SETTING UP PYTHON ENVIRONMENT AND LIBRARIES – JUPYTER NOTEBOOK

AIM:

To install Python, configure Jupyter Notebook, understand the notebook environment, execute Python programs, create Markdown cells, and use basic Jupyter Notebook features for Data Analytics applications

ALGORITHM:

1. Start the program.
2. Import the required libraries (if needed).
3. Declare and initialize the required variables or data.
4. Perform the required operations or processing.
5. Display the output/results.
6. Stop the program.

PROGRAM:

```
In [1]: #To print
        print("Welcome to Data Analytics Lab")
```

Welcome to Data Analytics Lab

```
In [2]: #declare variable and perform mathematical functions
        a = 20
        b = 10
        print(a+b)
        print(a-b)
        print(a*b)
        print(a/b)
```

30
10
200
2.0

```
In [3]: #declare variables and print it  
name = "John"  
dept = "AI&DS"  
cgpa = 8.9  
  
print("Name:",name)  
print("Department:",dept)  
print("CGPA:",cgpa)
```

Name: John
Department: AI&DS
CGPA: 8.9

```
In [4]: #import numpy and perform mean and sum operations  
import numpy as np  
a = np.array([10,20,30,40,50])  
print(a)  
print(np.mean(a))  
print(np.sum(a))
```

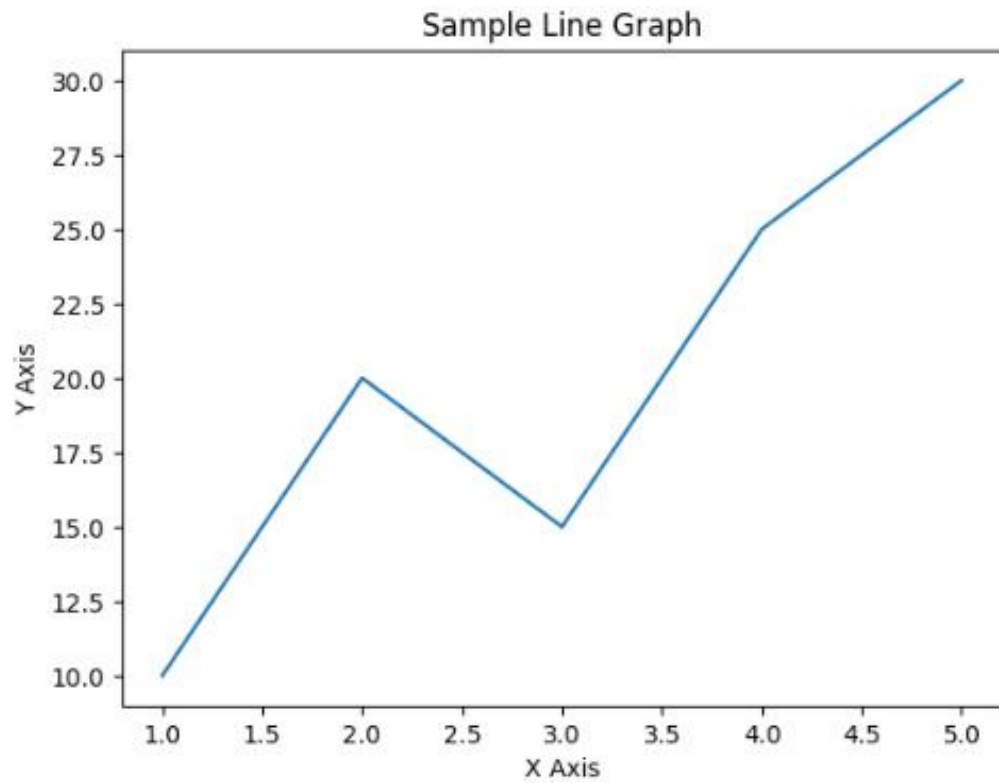
[10 20 30 40 50]
30.0
150

```
In [5]: #import pandas as pd and create a dcitionary  
#and make it as a DataFrame  
import pandas as pd  
data = {  
    'Name': ['John', 'David', 'Alex'],  
    'Marks': [85, 90, 95]  
}  
df = pd.DataFrame(data)  
print(df)
```

	Name	Marks
0	John	85
1	David	90
2	Alex	95

```
In [6]: #To plot a simple line graph
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[10,20,15,25,30]

plt.plot(x,y)
plt.title("Sample Line Graph")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.show()
```

**RESULT:**

The program to perform basic mathematical operations is completed successfully.